University of Geno – September 10, 2021
Mobile Robots Exam – 3rd session – 1 h 30 min

Exercise 1. Robot parameters (10 min)

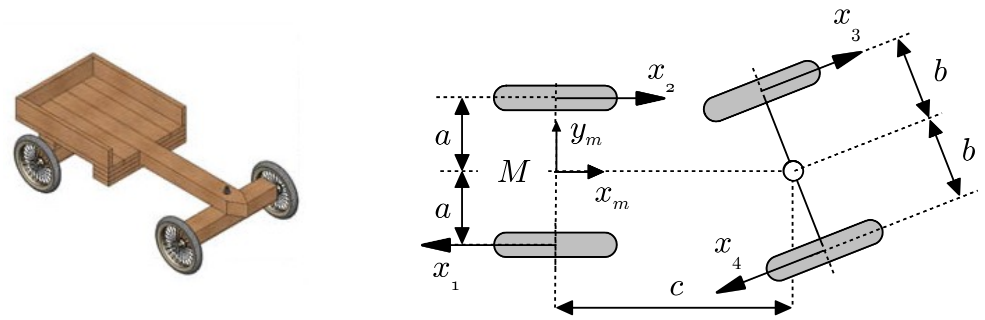


Figure 1.1. Robot to parameterize: the "soapbox"

Wheels 3 et 4 can be represented as two general wheels. In the parameter table, the fact that the wheels are rigidly linked will be reflected by **replacing β_4 by its expression as a function of β_3**. Give the parameter table of the robot.

Exercise 2. Kinematic modeling (45 min)

We are going to study the mobile robot defined by the following table of parameters:

W	L	α	d	β	γ	φ	ψ
	$a/2$	$\pi/2$	0	0	$\pi/2$		
	$a/2$	$-\pi/2$	0	0	$\pi/2$		
	b	0	0	β_{3s}	0		

Table 2.1. Parameters of the robot

In this exercise, failure to answer question 4.1. does not prevent from doing the rest of the exercise.

1. Give the schematic representation of the robot. The figure should show frame R_m, vectors x_1, x_2 and x_3, the dimensions a and b, and the angle β_{3s}. Make sure β_{3s} is indicated by a circular arrow with the origin at the zero position of the angle. I recommend you first draw on scrap paper and copy a **clean result** on your answer sheet once you are sure of your result. Clarity and cleanliness of the figure will be evaluation criteria.

2. Add the missing information of table 2.1. Define the configuration vector of the robot.

3. Calculate matrices J_1, J_2, C_1 and C_2 of the robot.

4. Define matrix C_1^* and **calculate** its rank. The rank of C_1^* must be **clearly and rigorously** established. A mere statement of the rank is not enough. It is **simple and fast**, but **rigorous** mathematical proof of the rank is required.

5. Give the kernel of C_1^*. Again, it is simple and fast but a mere statement of the kernel is not enough if it is not rigorously established.

6. Calculate the configuration kinematic model.

Exercise 3. Observability diagnostic without calculations (15 min).

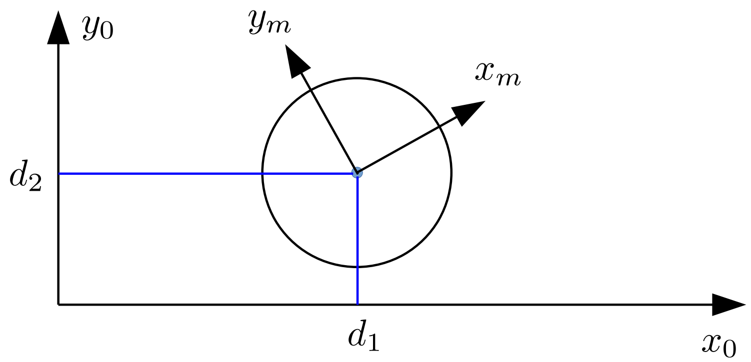


Figure 3.1. Omnidirectional robot and sensor

The **omnidirectional** robot of figure 1 is equipped with a sensor which measures the distances from its reference point M to two walls, which coincide with axes x_0 and y_0 of the absolute reference frame. The measurements are always taken perpendicular to the walls. How this is achieved is not your concern here.

The state is the posture $X = [x, y, \theta]^t$ of the robot. Explain, **without any calculation or reference to gradients or Lie derivatives**, why the state of the posture is never observable.

Recall: The posture kinematic model of the omnidirectional robot is $\dot{X} = U$.

A correct understanding of the definition of observability is the only prerequisite to solve this question.

Ten lines of text are more than enough for a complete and sound explanation. Take the time to think, and write only when you have something clear. Clarity and conciseness of the explanations are fundamental.

Exercise 4. Observability analysis by exact calculations (20 min).

A (2,0) robot is equipped with a sensor which measures **squared** distances from its reference point M to two walls. The walls are axes x_0 and y_0 of the reference frame (figure 4.1). The measurements are always taken perpendicular to the walls. How this is achieved is not your concern here.

The state is the posture $X = [x, y, \theta]^t$ of the robot. You have to prove that the **only** case when the state is not observable is $v = 0$. You will proceed by studying the determinant of sub-matrices of the observability matrix. To prove the result, you need to show that, for any other situation, there is at least one sub-matrix of the observability matrix which has a non zero determinant.

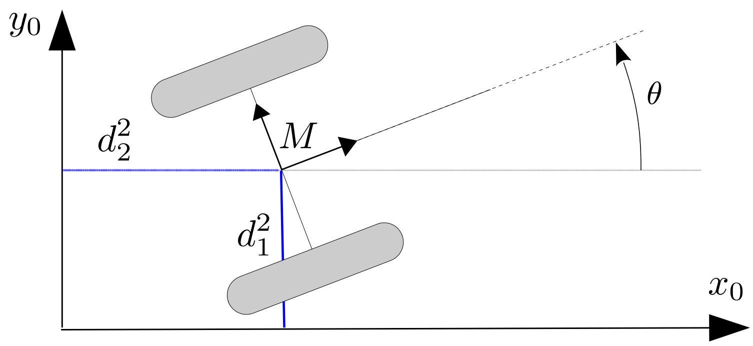


Figure 4.1. The robot, its sensor at point M and the measurements

Exercise 5. Equations of a Kalman filter (20 min).

The problem studied in this exercise is the localization of a (2,0) robot. Because of a high degree of slippage of the wheels, odometry cannot be used. Instead, the following two sensors are used to replace encoders:

- An accelerometer provides the acceleration γ of the vehicle which, by integration, will give the speed v.
- A gyrometer provides the rotational speed ω of the robot.

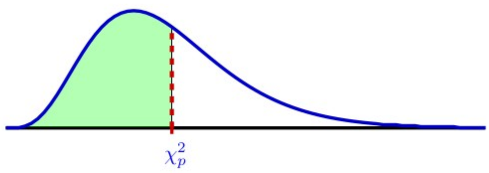
The posture of the robot is denoted by the usual notations of the class material: x, y, θ. The following equations relate the defined variables, in discrete form:

$x_{k+1} = x_k + v_k * \cos\theta_k$
$y_{k+1} = y_k + v_k * \sin\theta_k$
$\theta_{k+1} = \theta_k + \Delta\theta_k$
$v_{k+1} = v_k + \Delta v_k$

Where $\Delta v_k =_k *T$ and $\Delta\theta_k = \omega_k * T, T$ being the sampling period.

In addition, the robot is equipped with a GPS sensor which measures the position of the reference point M of the robot in the absolute refence frame in which we need to localize the robot.

1. Define a suitable state vector X, input vector U, evolution equation f and measurement equation g to solve the localization problem using a Kalman filter.
2. Calculate the matrices A, B and C of the Kalman filter.
3. The GPS is subject to incorrect measurements, often due to multi-paths (the wave from the satellite to the antenna does not follow a straight line: instead, it reaches the antenna after reflection on some surface, e.g. a wall, causing the position given by the GPS to be inaccurate). To protect the localization system from such erroneous measurements, a coherence test using the Mahalanobis distance is used. Using the χ^2 table below, define the Mahalanobis distance threshold that you would use in the test. Rapidly justify your choice.



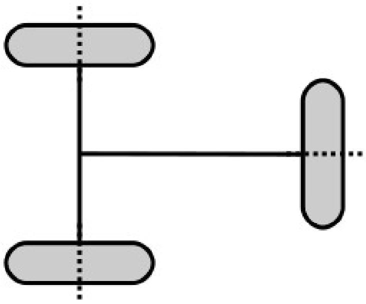
df	$\chi^2_{0.005}$	$\chi^2_{0.01}$	$\chi^2_{0.025}$	$\chi^2_{0.05}$	$\chi^2_{0.1}$	$\chi^2_{0.9}$	$\chi^2_{0.95}$	$\chi^2_{0.975}$	$\chi^2_{0.99}$	$\chi^2_{0.995}$
1	.0000	.0002	.0010	.0039	.0158	2.706	3.841	5.024	6.635	7.879
2	.0100	.0201	.0506	.1026	.2107	4.605	5.991	7.378	9.210	10.60
3	.0717	.1148	.2158	.3518	.5844	6.251	7.815	9.348	11.34	12.84
4	.2070	.2971	.4844	.7107	1.064	7.779	9.488	11.14	13.28	14.86
5	.4117	.5543	.8312	1.145	1.610	9.236	11.07	12.83	15.09	16.75

Figure 5.1. χ^2 table

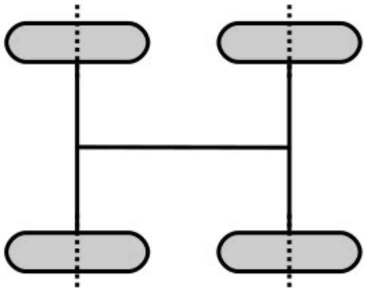
Exercise 6. Kinematics related short questions (10 min).

Do not copy the figures on your sheets. Just put the question number and the answer. When no justification is required, do not lose time giving any.

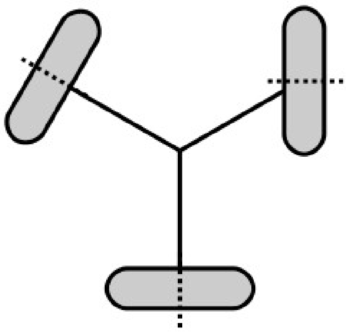
6.1. What is the degree of mobility of this wheel configuration? No justification required.



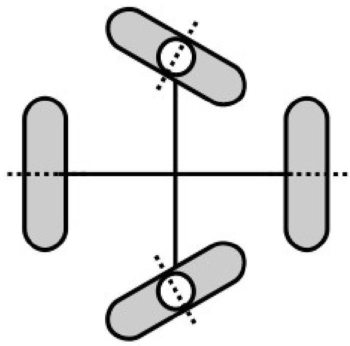
6.2. What is the degree of mobility of this wheel configuration? No justification required.



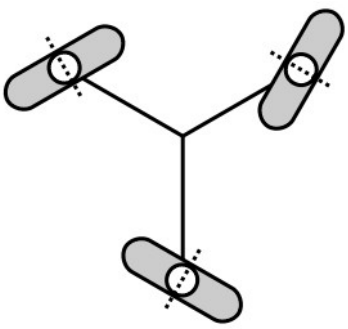
6.3. What is the degree of mobility of this wheel configuration? No justification required.



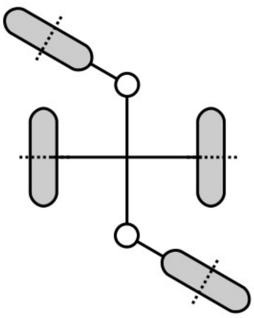
6.4. What is the type of this robot? No justification required.



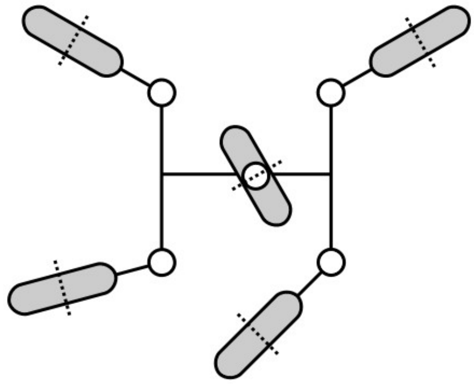
6.5. What is the type of this robot? No justification required.



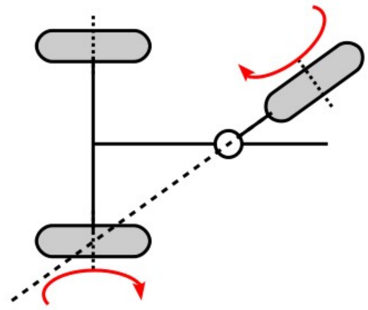
6.6. What is the type of this robot? No justification required.



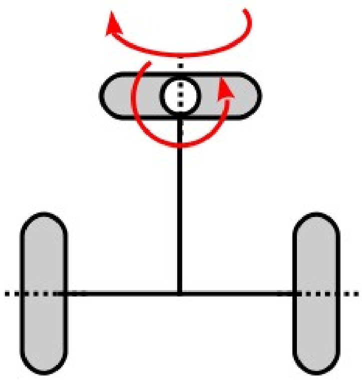
6.7. What is the type of this robot? No justification required.



6.8. The figure shows all the actuators of the robot. Is the configuration in the figure an actuation singularity? Shortly explain why (5 lines of text maximum). Your explanation must be short and clear.



6.9. Justify without any calculations that the following actuator configuration has no singularity (5 lines maximum).



6.10. Can you tell where the ICR is? If yes, at which point? If you think the ICR cannot be determined, say it explicitly. No justification is required.